EECE 430 — Software Engineering

Final Examination — Practice Exam 3

Name: ______________________________ Date: ____________________

Student ID: ___________________________ Section: __________________

Instructions:

- This exam contains **33 multiple-choice questions**.
- Time allowed: **1 hour**.
- Circle the letter of the **best** answer for each question.
- Each question is worth equal marks.
- No electronic devices, notes, or textbooks are permitted.

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1. Which agile principle states that the skills of the development team should be recognized and team members should develop their own ways of working?

A) Customer involvement
B) Incremental delivery
C) People not process
D) Maintain simplicity

2. A team practices TDD and all tests pass after implementing a feature. What should the team do NEXT before moving to the next feature?

A) Write additional documentation
B) Refactor the code to improve its structure while keeping all tests passing
C) Delete the tests since they are no longer needed
D) Present the feature to the Product Owner for approval

3. Which of the following is a benefit of test automation in agile development?

A) It eliminates the need for developers to understand the code
B) Tests can be scheduled to execute unattended overnight and analysis done the next morning
C) It removes the need for acceptance testing
D) It guarantees zero bugs in production

4. Incremental development in agile requires the team to prioritize features so that:

A) The least important features are implemented first to get them out of the way
B) All features are implemented simultaneously across parallel teams
C) The most important features are implemented first, with details defined only for the current increment
D) Features are randomly selected for each increment

5. Why might architecture refactoring be problematic in agile development even though code refactoring is encouraged?

A) Architecture refactoring is always cheaper than code refactoring
B) Architecture changes are much more expensive because they affect many components in the system
C) Agile forbids any changes to the architecture
D) Architecture is not relevant in agile projects

6. During a sprint review, the Product Owner determines that the sprint goal was NOT achieved. What is the appropriate response?

A) Fire the ScrumMaster
B) Extend the sprint until the goal is met
C) Confirm that the PBI implementation is incomplete, reflect on what went wrong, and carry items to future sprints

D) Skip the next sprint to compensate

7. Which XP practice ensures that no single developer becomes the sole expert on any part of the codebase?

A) Simple design
B) Sustainable pace
C) Collective ownership — all pairs work on all areas so no islands of expertise develop
D) Small releases

8. Story points differ from effort estimates expressed in person-hours because:

A) Story points are always more accurate than person-hour estimates
B) Story points are relative estimates that account for size, complexity, technology, and unknowns, while person-hours measure absolute time
C) Person-hours cannot be used in agile methodologies
D) Story points can only be used with the Fibonacci scale

9. What is the key difference between sprint planning output and product backlog?

A) They contain the same information in different formats
B) The sprint backlog is a breakdown showing what is involved in implementing the PBIs chosen for that sprint, while the product backlog lists all remaining work
C) The product backlog is created during the sprint and the sprint backlog is created at project start
D) Sprint planning produces the project budget while the product backlog produces the schedule

10. Which assumption of traditional Scrum may NOT hold in modern distributed teams?

A) Sprints should be timeboxed
B) The team will be co-located, sharing a workspace, with all members attending morning meetings
C) The product backlog should be prioritized
D) The development team should be self-organizing

11. A company is building a web application and must decide on an architecture. The system needs a browser-based UI, user authentication, application logic, shared services, and a database. Which architecture is MOST appropriate?

A) Pipe and Filter
B) A generic layered web architecture with UI, authentication, application, shared services, and database layers
C) A single monolithic executable with no layer separation
D) Repository pattern

12. What is the PRIMARY advantage of the client-server pattern when the load on a system is variable?

A) Clients can process data without any server
B) Servers can be replicated to handle increased load
C) The client-server pattern eliminates network dependency
D) All data is processed on the client side

13. An architect chooses a service-oriented architecture where all services are stateless. What key operational benefit does statelessness provide?

A) Services never need to access any data
B) Services can be replicated, distributed, and migrated from one server to another freely
C) Services do not need to communicate with each other
D) Services are immune to all security attacks

14. When choosing between relational and NoSQL databases for a system, which guideline from the course applies?

A) Always use NoSQL databases because they are newer
B) Relational databases like MySQL are suitable for transaction management; NoSQL databases like MongoDB are more flexible for data analysis

C) The database choice has no impact on architectural design
D) Both types are identical in capabilities

15. In a service-oriented architecture, what role does a service gateway play?

A) It stores all service data in a central repository
B) It acts as a single access point and proxy, enabling transformations, routing, and common processing so clients don't need to know communication protocols
C) It replaces the need for individual services
D) It compiles and deploys service code

16. A SaaS provider wants business customers to be able to authenticate users through their own corporate credentials rather than the provider's system. This customization is called:

A) Branding customization
B) Business rules customization
C) Authentication customization — delegating authentication to the customer's own identity system
D) Data schema customization

17. Which statement about scaling in cloud-based systems is CORRECT?

A) Scaling-up (using more powerful servers) is the preferred approach in cloud systems
B) Scaling-out (adding more virtual servers) is the normal approach, requiring software organized so components can be replicated and run in parallel
C) Cloud systems cannot be scaled once deployed
D) Scaling only applies to database servers, not application servers

18. A cloud system uses database mirroring where updates to the primary database are replicated to a standby database in a different location. This is an implementation of:

A) Scalability
B) Elasticity
C) Resilience — maintaining redundant copies so operation can switch automatically if the primary fails
D) Multi-tenancy

19. Which of the following is a disadvantage of SaaS for CUSTOMERS?

A) Try before you buy options
B) Continuous deployment of new versions
C) Loss of control over software updates — the provider decides when updates are applied
D) Reduced upfront costs

20. When extending a multi-tenant database using the 'additional tables' approach instead of additional columns, what is the main drawback?

A) It prevents tenants from having custom fields
B) It adds complexity to the database management software
C) It requires separate physical databases per tenant
D) It eliminates the ability to query data

21. In a centralized version control system, what mechanism prevents two developers from overwriting each other's changes to the same component?

A) The system automatically merges all changes without notification
B) When a component is checked out, the system warns other users, and upon check-in, the CMS ensures changes do not conflict
C) Only one developer can access the repository at a time
D) Developers must verbally coordinate who works on which files

22. Which of the following is a feature of code management systems that ensures files are never overwritten by later changes?

A) Storage management
B) Project support

C) Independent development — several developers can work on the same file and a new version is created upon submission

D) Change history recording

23. A DevOps team wants to ensure their testing environment is exactly the same as their deployment environment. Which practice BEST achieves this?

A) Manual configuration of each server by system administrators

B) Infrastructure as Code with Docker containers — using the same container image for both test and production

C) Writing detailed documentation about server setup

D) Hiring separate teams for test and production environments

24. According to the code completeness checklist in Scrum/DevOps, code is considered complete when it has been:

A) Written and compiled successfully

B) Reviewed by another team member, unit/integration tested, integrated with no errors, and accepted via acceptance tests

C) Committed to the repository by the developer

D) Demonstrated at the daily scrum meeting

25. What is the key difference between Git (the tool) and GitHub (the platform)?

A) They are the same thing

B) Git is a distributed version control system for tracking source code changes; GitHub is a cloud-based hosting service for managing Git repositories

C) GitHub is a programming language; Git is an IDE

D) Git only works offline; GitHub only works online

26. Which of the following scenarios BEST illustrates the 'cascading errors' disadvantage of microservices?

A) A developer writes a bug in one microservice that only affects that service

B) Service A calls Service B, which calls Service C. When Service C fails, Service B fails, which causes Service A and all applications using A to fail

C) Two microservices are deployed on the same container

D) A microservice's database grows too large

27. In the context of microservice data design, why should data sharing between services be minimized and preferably read-only?

A) Because microservices cannot use databases

B) To reduce the need for data consistency management — if services share writable data, changes must propagate to all services, adding overhead and complexity

C) Because read-only data is faster to encrypt

D) To prevent microservices from communicating at all

28. A system uses indirect (message broker) communication between microservices. Which of the following is a disadvantage of this approach compared to direct communication?

A) It makes it harder to replace individual services

B) It does not support asynchronous interactions

C) The overall system becomes more complex, and the broker can become a bottleneck or single point of failure

D) Services must know each other's exact network addresses

29. Why is the microservices approach described as having an 'increased security attack surface'?

A) Microservices use weaker encryption algorithms

B) Each microservice communicates over the network, creating multiple points that an attacker can target

C) Microservices cannot implement authentication

D) Microservices store all data in a single unprotected database

30. In a buffer overflow injection attack, what happens?

A) The attacker sends too many HTTP requests to overwhelm the server
B) Malicious code is injected through a valid input field, overflowing the allocated memory buffer and potentially executing attacker-controlled instructions
C) The attacker steals cookies from the user's browser
D) The attacker guesses the user's password through repeated attempts

31. A company implements federated identity (e.g., 'Login with Google'). Which of the following is an advantage of this approach for the USER?

A) The user must create a new unique password for every service
B) A single set of credentials is stored by a trusted identity service, reducing the number of passwords to track and the risk of breaches
C) The user's data is automatically shared with all federated services
D) The user no longer needs any form of authentication

32. According to data protection principles in the course, what must happen when a user deletes their account from a software product?

A) The data should be archived for future marketing use
B) The personal data associated with that account must be deleted — data should not be kept longer than needed
C) The data should be transferred to another user
D) The data should be encrypted but kept indefinitely

33. DevSecOps automates the integration of security at which phase(s) of the software development lifecycle?

A) Only during the testing phase
B) Only during deployment and delivery
C) In ALL phases: design, implementation, testing, deployment, and delivery
D) Only during the initial design phase

— END OF QUESTIONS —

EECE 430 — Practice Exam 3 — Answer Key

Quick Reference:

Q	Ans	Q	Ans	Q	Ans
1.	C	2.	B	3.	B
4.	C	5.	B	6.	C
7.	C	8.	B	9.	B
10.	B	11.	B	12.	B
13.	B	14.	B	15.	B
16.	C	17.	B	18.	C
19.	C	20.	B	21.	B
22.	C	23.	B	24.	B
25.	B	26.	B	27.	B
28.	C	29.	B	30.	B
31.	B	32.	B	33.	C

Detailed Explanations:

1. C — 'People not process' means recognizing the team's skills and letting them develop their own ways of working without prescriptive processes.

2. B — In TDD, after all tests pass, the next step is to refactor the code for improvement. Then move on to the next feature increment.

3. B — Test automation benefits include faster execution, greater coverage, less human resources needed, and the ability to run tests unattended (e.g., overnight).

4. C — Agile prioritizes features so the most important ones are built first. Only the details of the feature being implemented in the current increment are defined.

5. B — Architecture refactoring is much more expensive than code refactoring because architectural changes ripple through many system components.

6. C — The Product Owner confirms whether the goal is met. If not, the team reflects on what went wrong, and incomplete items return to the backlog.

7. C — Collective ownership means pairs of developers work on all areas of the system, preventing knowledge silos and ensuring shared responsibility.

8. B — Story points are relative, factoring in size, complexity, technology, and unknowns. Person-hours are absolute time estimates. Story points are estimated by comparison with a baseline task.

9. B — Sprint planning produces a sprint backlog — a detailed breakdown of PBIs for that sprint. The product backlog is the full prioritized list of all remaining work.

10. B — Scrum assumes co-located, full-time teams attending daily morning meetings. In reality, teams may be distributed, part-time, or on flexible hours.

11. B — The course describes a generic layered web architecture with exactly these layers: browser UI, authentication/UI management, application logic, shared services, and database/transaction management.

12. B — Client-server is used when load is variable because servers can be replicated across a network to handle demand.

13. B — Stateless services can be replicated and migrated freely because they hold no internal state. State is stored in a database or passed with each request.

14. B — The course states that relational databases (MySQL) suit transaction management, while NoSQL (MongoDB) is more flexible for data analysis.

15. B — A service gateway is a single access point acting as a proxy. It handles transformations, routing, and common processing across services.

16. C — Authentication customization allows businesses to use their own corporate credentials (federated/delegated authentication) instead of the SaaS provider's credentials.

17. B — In cloud systems, scaling-out (adding servers) is preferred over scaling-up. Software must be organized so individual components can be replicated and parallelized.

18. C — Database mirroring with automatic failover is a resilience mechanism — redundant copies in different locations ensure continuity if the primary fails.

19. C — Customers lose control over when updates happen — the SaaS provider pushes updates to all customers simultaneously.

20. B — Using separate extension tables for custom fields adds complexity to the database management software compared to the simpler additional columns approach.

21. B — In centralized VCS, checked-out files are marked as shared. The system warns others and ensures no conflicts on check-in by comparing versions.

22. C — Independent development: multiple developers can work on the same file simultaneously. New versions are created on submission so files are never overwritten.

23. B — Docker containers and IaC ensure identical environments. Using the same container image for test and production guarantees they are the same.

24. B — The checklist requires: reviewed (meets coding standards), unit/integration tested (all pass), integrated (no errors), and accepted (acceptance tests pass, PO confirms).

25. B — Git is the DVCS tool itself. GitHub is a cloud platform that hosts Git repositories and provides collaboration features on top of Git.

26. B — Cascading errors: one service invokes another in a chain. If one fails, the failure propagates back through all calling services and their dependent applications.

27. B — Minimizing shared writable data reduces the burden of data consistency management. Read-only sharing avoids propagation overhead from updates across services.

28. C — Message brokers add system complexity and can become bottlenecks or single points of failure, though they decouple services and support easier modification.

29. B — Each microservice communicates via network messages, and each communication point is a potential attack vector — more services means more targets.

30. B — Buffer overflow: malicious input exceeds allocated memory, potentially allowing the attacker to execute injected code.

31. B — Federated identity means one set of credentials at a trusted provider. Fewer stored passwords means less breach exposure and less for users to remember.

32. B — The 'data lifetime' principle: you must not keep data longer than needed. When a user deletes their account, their personal data must be deleted.

33. C — DevSecOps automates security integration across ALL phases: design, implementation, testing, deployment, and delivery.

— END OF ANSWER KEY —